SAKARYA UNIVERSITY · COP31 GREEN ZONE ANTALYA 2026

yeşilkod

Measures software energy, fixes the waste, proves the gain.

How much electricity does this line of code use?

```
- return [no for no in GELENLER if no in kayitlilar]
+ kayitlilar_kume = set(kayitlilar)
+ return [no for no in GELENLER if no in kayitlilar_kume]
```

465× **less energy, identical output.**

At one million requests a day, this single line emitted about 41 kg CO_2e a year. The engine rewrote it by itself.

WHY IT MATTERS

Code has an invisible carbon cost. Yeşil Kod makes it visible.

Data-centre electricity use is expected to grow from **415 TWh** in 2024 to **~945 TWh** in 2030 (IEA). Part of that energy feeds waste nobody notices: loops that repeat the same work thousands of times, needless database queries, results computed over and over.

1 Find
Walks the syntax tree and flags nine energy-wasting patterns, line by line.

2 Fix
Rewrites four of them itself: batched query, set lookup, loop hoisting, caching.

3 Verify
Runs both versions and rejects the fix unless the outputs are identical.

4 Measure
Reads CPU energy counters and converts the gain to carbon on the Türkiye grid.

Green Coding and Digital Carbon Footprint Optimisation Tool
"Yeşil Kod" is Turkish for "Green Code".

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RESULTS

Four real-world wastes. The engine fixed all four.

Route count	uncached double recursion → lru_cache	**~4,000×**
Attendance check	list lookup inside a loop → set()	**465×**
Price list	loop-invariant work → moved out of the loop	**51×**
Order report	N+1 queries → one batched IN query	**4,001 → 2 queries**

Reduction in CPU energy per request. Estimated measurement mode, 26 October 2026. None of the optimised code was written by hand.

Carbon accounting

SCI = (E × I + M) / R

- **E** measured CPU energy
- **I** Türkiye grid, 341 gCO_2e/kWh
- **M** embodied emissions (next phase)
- **R** per request

Honest limits

- Only CPU energy is measured; memory, network and cooling are excluded. The gain is a lower bound.
- Caches are cleared before every request.
- The engine currently supports Python.

SCAN ME

Try your own code on your phone.

The analysis runs on your phone; your code never leaves it. The site also has the film with English subtitles.

→ yes3760.github.io/yesil-kod
→ Source: github.com/yes3760/yesil-kod

Sakarya University · Team: Yunus Emre Sevindik · Mehmet Zahid Görgeç
Sources: IEA, Energy and AI (2025) · Low-Carbon Power, Türkiye (Aug 2025–Jul 2026) · Green Software Foundation, SCI

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